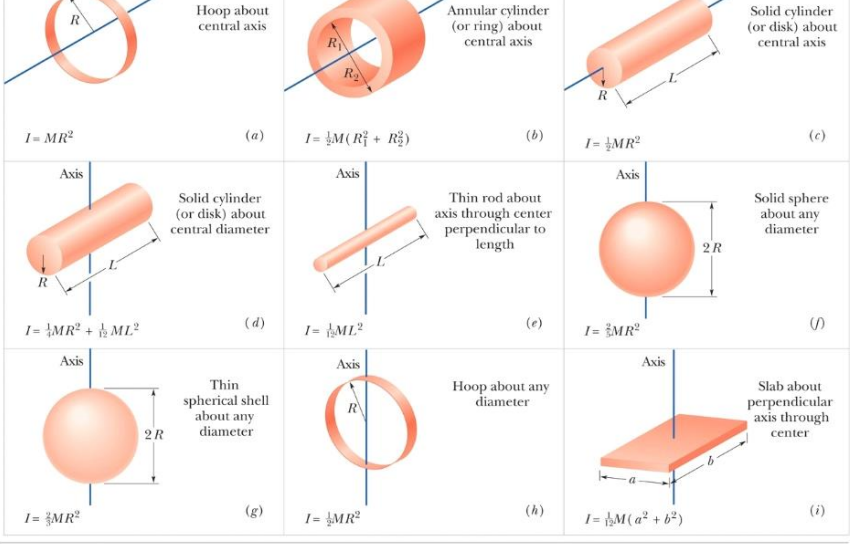


COM of a system

- For solid bodies, we take the limit of an infinite sum of infinitely small particles → integration!

$$x_{com} = \frac{1}{M}\int x\,dm,\ y_{com} = \frac{1}{M}\int y\,dm,\ z_{com} = \frac{1}{M}\int z\,dm$$

- For Uniform Objects

$$\rho = \frac{dm}{dV} = \frac{M}{V}$$

$$x_{com} = \frac{1}{V}\int x\,dV,\ y_{com} = \frac{1}{V}\int y\,dV,\ z_{com} = \frac{1}{V}\int z\,dV$$

- COM (uniform objects) lies at a point (line/plane) of symmetry (if there is any), which needs not lie within the object.

COM of a system

- Many Particles

$$x_{com} = \frac{m_1x_1 + m_2x_2 + m_3x_3 + \cdots + m_nx_n}{M} = \frac{1}{M}\sum_{i=1}^{n} m_ix_i$$

- Three dimensions

$$x_{com} = \frac{1}{M}\sum_{i=1}^{n} m_ix_i,\ y_{com} = \frac{1}{M}\sum_{i=1}^{n} m_iy_i,\ z_{com} = \frac{1}{M}\sum_{i=1}^{n} m_iz_i$$

$$\vec{r}_{com} = x_{com}\hat{\imath} + y_{com}\hat{\jmath} + z_{com}\hat{k}$$

$$\vec{r}_{com} = \frac{1}{M}\sum_{i=1}^{n} m_i\vec{r}_i$$

Projectile Motion

Effects of Air have been eliminated

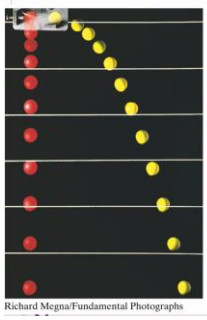
- Projectile Motion: A particle moves in a vertical plane with initial velocity $\vec{v}_0$, but its acceleration is always the free-fall acceleration a= $-g$
- In projectile motion, the horizontal motion and the vertical motion are independent of each other; that is, neither motion affects each other

$$\vec{v}_0 = v_{0x}\hat{\imath} + v_{0y}\hat{\jmath}$$

$$v_{0x} = v_0\cos\theta_0 \text{ and } v_{0y} = v_0\sin\theta_0$$

The $\vec{r}$ and $\vec{v}$ change continuously, but $\vec{a}$ is constant and always pointing vertically downward.

Projectile Motion



- Independent Motion➔ two separate 1D problem
- Y-axis

$$y - y_0 = (v_0\sin\theta_0)t - \frac{1}{2}gt^2$$

$$v_y = v_0\sin\theta_0 - gt$$

$$v_y^2 = (v_0\sin\theta_0)^2 - 2g(y - y_0)$$

- X-axis (Trajectory)

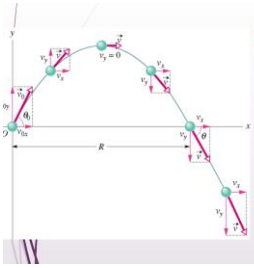
$$x - x_0 = (v_0\cos\theta_0)\,t$$

- X-Y relationship

$$y = (\tan\theta_0)x - \frac{gx^2}{2(v_0\cos\theta_0)^2}$$

- x_0 and y_0 are zero.
- θ_0 is the angle between $\vec{v}$ and +ve $\hat{x}$.

ige



- The horizontal distance the particle has traveled when it returns to its initial height, when y-y_0=0

$$R = (v_0\cos\theta_0)\,t = \frac{v_0^2}{g}\sin 2\theta_0$$

- R_{max}, when $\theta_0 = 45°$

We have used the relation $\sin 2\theta_0 = 2\sin\theta_0\cos\theta_0$

The Maximum Height

- The maximum height the particle can reach during the projectile motion as shown in previous slide, $v_y = 0$

$$v_y = v_0\sin\theta_0 - gt$$

$$t_h = \frac{v_0\sin\theta_0}{g}$$

$$y - y_0 = (v_0\sin\theta_0)t - \frac{1}{2}gt^2$$

$$H_{max} = y - y_0 = \frac{(v_0\sin\theta_0)^2}{2g}$$

$$R = \frac{2v_0^2}{g}\sin\theta_0\cos\theta_0$$

$$\frac{H_{max}}{R} = \frac{\tan\theta_0}{4}$$

The relation between the maximum height and horizontal range

Projectile Motion

$$y - y_0 = (v_0\sin\theta_0)t - \frac{1}{2}gt^2$$

$$v_y = v_0\sin\theta_0 - gt$$

$$v_y^2 = (v_0\sin\theta_0)^2 - 2g(y - y_0)$$

$$x - x_0 = (v_0\cos\theta_0)\,t$$

$$y = (\tan\theta_0)x - \frac{gx^2}{2(v_0\cos\theta_0)^2}$$

周期 T: The **period of revolution** or simply **period**

$$x_{PA} = x_{PB} + x_{BA}$$

The coordinate x_{PA} of *P* as measured by *A* is *equal to* the coordinate x_{PB} of *P* as measured by *B plus* the coordinate x_{BA} of *B* as measured by *A*. Note that x is a vector in one dimension.

$$\frac{d}{dt}x_{PA} = \frac{d}{dt}x_{PB} + \frac{d}{dt}x_{BA}$$

$$v_{PA} = v_{PB} + v_{BA}$$

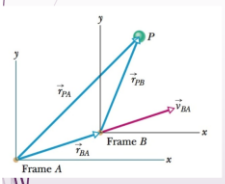
- The velocity v_{PA} of *P* as measured by *A* is *equal to* the velocity v_{PB} of *P* as measured by *B plus* the velocity v_{BA} of *B* as measured by *A*. The term v_{BA} is the velocity of frame B relative to frame A.
- If v_{BA} is constant, which means B is driving at a constant speed relative to A.

$$\frac{d}{dt}v_{PA} = \frac{d}{dt}v_{PB} + \frac{d}{dt}v_{BA}$$

$$a_{PA} = a_{PB} + a_{BA} = a_{PB}$$

- The acceleration a_{PA} of *P* as measured by *A* is *equal to* the acceleration a_{PB} of *P* as measured by *B plus* the acceleration a_{BA} of *B* as measured by *A*.
- Observers on different frames of reference that move at constant velocity relative to each other will measure the same acceleration for a moving particle. (Inertia Frames)

Relative Motion in Two Dimensions



- Two observers A and B are watching moving particle P from the origins of reference frames A and B, while B moves at a constant velocity $\vec{v}_{BA}$ relative to A.

$$\vec{r}_{PA} = \vec{r}_{PB} + \vec{r}_{BA}$$

$$\vec{v}_{PA} = \vec{v}_{PB} + \vec{v}_{BA}$$

$$\vec{a}_{PA} = \vec{a}_{PB}$$

Observers on different frames of reference that move at constant velocity relative to each other will measure the same acceleration for a moving particle. (Inertia Frames)

Free-body Diagram

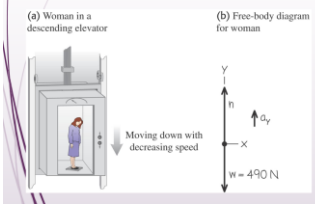
To solve the problem with Newton's Second Law

Free-body diagram

- In a coordinate system, we represent the body as a dot, each force on the body is a vector arrow with its tail on the body.
- A collection of two or more bodies is called a system, and any force on the bodies inside the system caused by external agents outside the system is called an external force. If the bodies are rigidly connected, then we can treat the system as one composite body, and the net force $\vec{F}$ on it is the vector sum of all external forces.

Weighing

A woman inside the elevator is standing on a scale. How will the acceleration of the elevator affect the scale reading?



$\vec{a}_y$: +ve Y-axis
$W - mg = ma_y$
$W = m(g + a_y)$

$\vec{a}_y$: -ve Y-axis
$W - mg = -ma_y$
$W = m(g - a_y)$

Drag Force ($\vec{D}$) 拖拽力

Def.: When there is a relative velocity between a fluid and a body, the body experiences a drag force.

Direction: Opposes the relative motion and points in the direction in which the fluid flows relative to the body.

$$D = \frac{1}{2}C\rho A v^2$$

C: drag coefficient

(C: Typical values range from 0.4 to 1.0; and we use it as constant here.)

ρ: air density

A: effective cross-sectional area

(perpendicular to the velocity)

v: relative speed

(Drag force 是指当一个物体相对于周围流体运动时，与其相反的作用力。这种力可以存在于两个流体层（或表面）之间，也可以存在于流体和固体表面之间。当物体在流体中移动时，流体会产生阻力，称为拖曳力或拖曳。它与物体的运动方向相反，并由固体物体和流体之间的相对速度产生。)

-Terminal Speed (v_t) 终端速度

Def.: When a blunt body falls from rest through air, If |D| = |mg|, then the body's speed is constant (the acceleration a=0),

which is called **terminal speed v_t**

$$D - F_g = ma$$
$$\frac{1}{2}C\rho A v_t^2 - F_g = 0$$
$$v_t = \sqrt{\frac{2F_g}{C\rho A}}$$

(a=0)

-Uniform Circular Motions

Centripetal force 向心力

Magnitude: (F=ma, Magnitude of the acceleration $a = \frac{v^2}{r}$)

The direction: Pointing to the center of the circle all the time

- Acceleration

$$\vec{a} = \frac{d\vec{v}}{dt}$$

$$\frac{dy_p}{dt} = v_y = v\cos\theta\,; \quad \frac{dx_p}{dt} = v_x = -v\sin\theta$$

$$= \left(-\frac{v}{r}\frac{dy_p}{dt}\right)\hat{\imath} + \left(\frac{v}{r}\frac{dx_p}{dt}\right)\hat{\jmath}$$

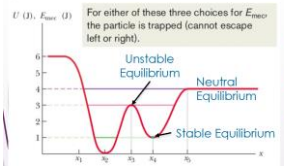
$$= \left(-\frac{v^2}{r}\cos\theta\right)\hat{\imath} + \left(-\frac{v^2}{r}\sin\theta\right)\hat{\jmath}$$

$$a = \sqrt{a_x^2 + a_y^2} = \frac{v^2}{r}$$

$$\tan\phi = \frac{a_y}{a_x} = \tan\theta$$

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Potential Energy Curve



If the E_{mec} is
- Purple Line: after x_5, $K = 0$, Neutral Equilibrium
- Pink Line: at x_3, $K = 0$, Unstable Equilibrium
- Green Line: at x_4, $K = 0$, Stable Equilibrium

- Conservative Forces
- Potential Energy

$$\Delta U = -\int_{x_i}^{x_f} F(x)dx$$

- Gravitational Potential Energy

$$U(y) = mgy \ (+\text{ve: up})$$

- Elastic Potential Energy

$$U(x) = \frac{1}{2}kx^2$$

- Mechanical Energy

$$\Delta E_{mec} = \Delta K + \Delta U$$

- Potential Energy Curve

$$F(x) = -\frac{dU(x)}{dx}$$

- Turning Point, when K=0
- Neutral/Unstable/Stable Equilibrium
- Work Done on a System by an External Force

$$W = \Delta E_{mec} + \Delta E_{th}$$

- Conservation of Energy

$$W = \Delta E = \Delta E_{mec} + \Delta E_{th} + \Delta E_{int}$$

- Power

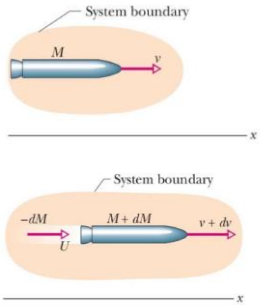
$$P_{avg} = \frac{\Delta E}{\Delta t}$$
$$P = \frac{dE}{dt}$$

ΔE_mec 机械能变化量、ΔE_th 热能变化量、ΔE_int 内能变化量

- Spring Force
  - $\vec{F_s} = -k\vec{d}$ or $F_x = F(x) = -kx$
  - Hooke's Law

-System with Varying Mass: A Rocket

The ejection of mass from the rocket's rear increases the rocket's speed.



(a)从惯性参考系看，在时间 t 处质量为 M 的加速火箭。

(b) 相同，但在时间 t+dt。图中显示了间隔 dt 期间释放的排气产物。- 火箭和排气产品形成一个孤立的系统

-Conserve Momentum

· $Mv = -dMU + (M + dM)(v + dv)$

U: velocity of products relative to frame

dM:-ve

Use relative speed（v_{rel}）: velocity of rocket relative to products

· $v + dv = v_{rel} + U$

$U = \mathbf{v} + \mathbf{dv} - \mathbf{v_{rel}}$

$-dMv_{rel} = Mdv$

-Rocket Equations

For rockets releasing products:

· $-\frac{dM}{dt}v_{rel} = M\frac{dv}{dt}$

R: Mass rate of fuel consumption $-\frac{dM}{dt} = R$;

$T = Rv_{rel} = Ma$

First Rocket Equation

R v_{rel} : the Thrust of the rocket engine (T)

· $dv = -v_{rel}\frac{dM}{M}$

$\int_{v_i}^{v_f} dv = -v_{rel}\int_{M_i}^{M_f}\frac{dM}{M}$

$v_f - v_i = v_{rel}\ln\frac{M_i}{M_f}$

The velocity change for a given consumption of fuel as the second rocket equation.

零角位置. · $\theta = \frac{s}{r}$ * θ 必须使用弧度制

· $\omega_{avg} = \frac{\theta_2 - \theta_1}{t_2 - t_1} = \frac{\Delta\theta}{\Delta t}$ ($\Delta\theta = \theta_2 - \theta_1$ 仅对刚体成立)

·$\omega = \lim_{\Delta t\to 0}\frac{\Delta\theta}{\Delta t} = \frac{d\theta}{dt}$

$\alpha_{avg} = \frac{\omega_2 - \omega_1}{t_2 - t_1} = \frac{\Delta\omega}{\Delta t}$对任意物体都成立

(!!! Right-Hand Rule 右手定则 用于 判断 角速度 ω 和 角加速度 α 的方向)

For fixed rotational axis,

only two ways of rotation:

Counterclockwise - Positive(+)

Clockwise - Negative(-)

For rotations about **different axes**: $\overrightarrow{\theta_1} + \overrightarrow{\theta_2} \neq \overrightarrow{\theta_2} + \overrightarrow{\theta_1}$

For **rotations about a fixed axis**: $\theta_1 + \theta_2 = \theta_2 + \theta_1$

Linear Equation	Angular Equation
$v = v_0 + at$	$\omega = \omega_0 + \alpha t$
$x - x_0 = v_0 t + \frac{1}{2}at^2$	$\theta - \theta_0 = \omega_0 t + \frac{1}{2}\alpha t^2$
$v^2 = v_0^2 + 2a(x - x_0)$	$\omega^2 = \omega_0^2 + 2\alpha(\theta -$
$x - x_0 = \frac{1}{2}(v_0 + v)t$	$\theta - \theta_0 = \frac{1}{2}(\omega_0 + \omega)t$
$x - x_0 = vt - \frac{1}{2}at^2$	$\theta - \theta_0 = \omega t - \frac{1}{2}\alpha t^2$

a_r: 线性加速度 的 径向分量

· $a_r = \frac{v^2}{r} = \omega^2 r$ (radian measure)

a_t: 线性加速度 的 切向分量

· $a_t = \frac{dv}{dt} = \frac{d\omega}{dt}r = \alpha r$ (radian measure)

旋转动能 $K = \sum\frac{1}{2}m_i v_i^2 = \frac{1}{2}(\sum m_i r_i^2)\omega^2 = \frac{1}{2}I\omega^2$

$I = \sum m_i r_i^2 = \int r^2\,dm$xis

(?? 此处 dm 可写为 ? dr 的形式，然后在对于 r 的积分上，下边是相对于旋转轴的负坐标，上边是相对于旋转轴的正坐标 ?)

-The Parallel-Axis Theorem 平行轴定理

Have known the moment of inertia for the center of mass axis (I_{com})

→ Find the moment of inertia for a parallel axis (I)

· $I = I_{com} + Mh^2$

h: the perpendicular distance between the axis that we need & the axis through the center of mass

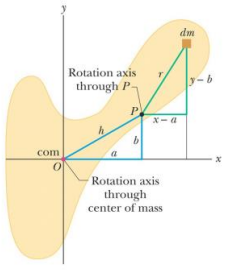
(these two axes must be parallel; not valid for two arbitrary axes)

M: 物体的总质量

I: 物体绕 任意轴(所需求 I 的轴) 的转动惯量

I_{com}: 物体绕 过其重心的轴 旋转时的转动惯量

We need to relate the rotational inertia around the axis at P to that around the axis at the com.



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- For Center of Mass:

$$x_{com} = \frac{1}{M}\sum_{i=1}^{n} m_i x_i = \frac{1}{M}\int x\,dm = 0$$

$$y_{com} = \frac{1}{M}\sum_{i=1}^{n} m_i y_i = \frac{1}{M}\int y\,dm = 0$$

- For Point at (x, y) rotating at axis-P

$$I = \int r^2\,dm = \int[(x-a)^2 + (y-b)^2]dm$$

$$= \int(x^2 + y^2)dm - 2a\int x\,dm - 2b\int y\,dm + \int(a^2 + b^2)\,dm$$

$$\int(x^2 + y^2)dm = \int R^2\,dm = I_{com}$$

$$\int(a^2 + b^2)\,dm = Mh^2$$

$$\vec{p}_1 = 4\hat{\imath} - 3\hat{\jmath} + 5\hat{k};$$
$$\vec{p}_2 = -6\hat{\imath} + 3\hat{\jmath} - 2\hat{k}.$$

$$\vec{p}_1 + \vec{p}_2 = -2\hat{\imath} + 0\hat{\jmath} + 3\hat{k}$$

$$\vec{p}_1 \times \vec{p}_2 = \begin{vmatrix} \hat{\imath} & \hat{\jmath} & \hat{k} \\ 4 & -3 & 5 \\ -6 & 3 & -2 \end{vmatrix} = \hat{\imath}(6-15) - \hat{\jmath}(-8+30) + \hat{k}(12-18)$$

$$= \hat{\imath}(-9) - 22\hat{\jmath} - 6\hat{k}$$

$$= -9\hat{\imath} - 22\hat{\jmath} - 6\hat{k}$$